DEEP LEARNING–BASED APPROACH FOR BRAIN TUMOR DETECTION USING MRI IMAGES

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Abstract—Early diagnosis and treatment planning for patients with tumors in the brain are very essential for better management. Magnetic Resonance Imaging (MRI) is a commonly used technique for analyzing images associated with a tumor in the brain; however, manual processing for diagnosis and interpretation is a complex and highly expert-intensive procedure for radiologists, sometimes even resulting in inconsistent diagnosis. In this scenario, to overcome this demerit, this paper presents a new concept for accurate diagnosis by a deep learning-based technique for analyzing MRI images.

The proposed architecture also incorporates Different types of deep learning models are used, such as Convolutional Neural Network (CNN) models for tumor classification, U-Net models, and Attention U-Net models are precise tumor segmentation. Moreover, The Super-Resolution Generative Adversarial Network (SRGAN) model is also used to increase the spatial resolution of the MRI images to capture the details within the tumor. The performance of the proposed architecture is tested using the publicly available brain MRI images. The performance measures include accuracy, Dice Coefficient, and Intersection over Union.

Experimental outputs show that the developed solution is effective for tumor detection and segmentation and outperforms standard CNN and U-NET methods. In particular, the method of tumor segmentation using the attention mechanism is capable of correctly outlining boundaries of the tumor. The experiments conducted confirm the efficiency of deep-learning methods for assisting doctors and increasing the reliability of computer-assisted diagnosis systems for brain tumor analysis.

Keywords— Brain Tumor Detection, MRI, Deep Learning, CNN, U-Net, Attention U-Net

I. INTRODUCTION

Tumors of the brain are among the most aggressive of central nervous system afflictions and may become dangerous to life if not identified promptly. The human brain is a very complex organ; therefore, the exact identification of brain tumors is essential for effective treatment planning that can help improve patient survival rates. Magnetic Resonance Imaging (MRI) is an important tool in testing brain tumors because it enables a clearer visualization of soft tissues than other modalities due to improved contrast. However, despite these advantages, MRI scan interpretation is labor-intensive, subjective, and highly dependent on the expertise of radiologists, which can lead to diagnostic inconsistencies.

Traditional image processing and classic machine learning use handcrafted texture features, among others; description of intensity and shape. These approaches show weaknesses in the representation of intricate spatial patterns and

contextual information in MRI brain images, which may show up as irregular shapes of tumors or their clearcut boundaries. Therefore, only large and heterogeneous data are suitable for the performance of such methods.

Recent Deep Learning developments have significantly improved the process of medical image analysis because they allow the acquisition of hierarchical features directly from raw data in an automatic fashion. Encoders, convolutional Neural Networks (CNN), and encoder- decoders such as U-Net have demonstrated very good results in detecting and segmenting brain tumors. Attention-based models further improve the accuracy of segmentation through the targeting of tumor- relevant contents while softening the irrelevant background. Finally, under the super-resolution category, one can mention Super- Resolution Generative Adversarial Networks - SRGAN - which are employed to enhance MRI images by increasing their spatial resolution and fine details.

The trends indicate that this paper will propose a Deep Learningbased model of automatic brain tumor recognition and segmentation with MRI images. It merges CNN, U-Net, Attention UNet, and SRGAN-based enhancement of images in order to capture proper localization and segmentation of the tumor. The suggested approach is tested on publicly obtained brain MRI data with standard performance measures.

II. RELATED WORK

The first research on brain tumor detection relied primarily on the traditional image processing and classical machine learning methods. These approaches normally entailed tedious extraction of features which comprised of texture, intensity and shape- based design of descriptors by domain specialists. They were then classified using support vectors machine, decision tree and random forest[11]. Although these methods provided sensible outcomes on small data sets, they did do not generalize well since they cannot necessarily reflect complex spatial patterns and changes that can be found in brain MRI images.

With the advancements in machine learning technologies, various approaches have been introduced in relation to the automatic classification of brain tumors, using the concepts of supervised machine learning techniques, with manually defined feature representations. The success of

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such approaches relied heavily on the quality of the features that were manually defined[12]. However, the variation in the size of the tumors, the irregular boundary, and the intensity variation in the patterns proved to be a challenge, thus limiting the accuracy of these approaches[13].

In the last two years, a new trend in the analysis of medical images using Deep Learning concepts has emerged. Convolutional Neural Network (CNN) architectures are also proved to be highly efficient in terms of automated learning of hierarchical features without the need for any manual feature extraction in their raw images i.e., without the need for any feature extraction. Other models such as the U-Net encoder decoder architecture also improved the efficiency of the segmentation process as they provided high localization with the assistance of skip connections that utilized both the low-level and high-level features. Their approach of U-Net was also utilized in attention in order to attend to the regions of interest in the tumor and to filter out the noise in the background regions that led to the depiction of the boundary demarcation to finer values[14].

In addition to the detection and segmentation models, image quality improvement tools were also taken into account in order to increase the diagnostic performance rates. It has been found that super-resolution algorithms, particularly, the Super- resolution Generative Adversarial Networks (SRGAN) are capable of improving the spatial resolution and look of the MRI images. The resulting high-quality feature learning and improved segmentation at the cost of low-quality input is brought about by these methods. Despite these advances combination of Super-resolution and strong segmentation architectures remain a research area, which drives the development of holistic approaches based on Deep Learning[15].

Alsheikhy, A. et al(2023)[16]: Recent trends and approaches to the design of an effective diagnosis system for detecting and categorizing brain tumors underscore the importance of integrating medical imaging with machine learning and deep learning concepts and ideas. Magnetic Resonance Imaging or MRI with the use of efficient preprocessing and feature extraction techniques has shown notable success in accurately segmenting tumors and detecting their locations within the brain. In categorizing cancer tumors within the brain, results clearly indicate the efficiency of deep learning models like the Convolutional Neural Network and its hybrid models in comparison to other classifiers.

Lin, D. J., et al.,(2023)[17]: It utilizes the acquisition techniques of multiple slices simultaneously, the compress-sensing concept, as well as denoising/upscaling processes that use the services of a neural network for restoring high-resolution TSE contrasts from very heavily undersampled data. The outcome of their experiments suggests that times for joint exams have been reduced to minutes, with no loss of resolutions.

Islam, M. M., et al.,(2023[18]): "Transfer learning with ImageNet-pretrained CNNs such as VGG, ResNet, DenseNet, MobileNet, and others has become the key to successfully solving the classification tasks for brain tumors in MRI images with high performance using small medical datasets." Transfer learning approaches to the MRI brain tumor classification problem have been shown to result in a higher quality of learning due to the application of fine-tuning methods compared to the direct use of feature extraction transfer learning.

III. PROPOSED METHODOLOGY

The proposed solution provides a Deep Learning system for identification and segmentation of brain tumors through an MRI image. The complete system relies on the raw images extracted from the MRI and is capable of providing high-quality images with relevant features and precise identification of the areas that contain tumors. The proposed system design relies on various Deep Learning models like CNN, U-Net, Attention-U-Net, and Super Resolution Generative Adversarial Network (SRGAN) that work together for improving the image quality. The working of the proposed solution is explained in Fig. 1. The images given out in publicly available datasets through MRI images contain variability in terms of resolution, intensity, and presence of noise.

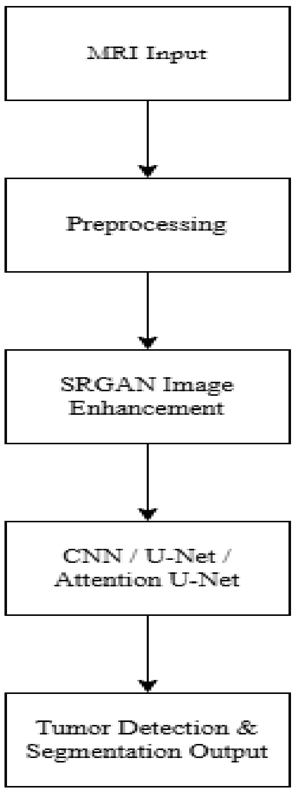

Fig. 1. Overall image processing flow with in the proposed brain tumor detection and segmentation system.

A. Data Preprocessing

This preprocessing might, therefore, be a major process that would give consistency and would make the functionality of models better. In this project, MRI images have undergone resizing processes to a fixed size and normalizing processes to make the intensity values normal. For enhancing the clarity within the tumor areas, there are noise reducing processes and contrast enhancement processes. This pre-processing work ensures that both the faster convergence rates of the model and learning from distinctive features are enhanced.

B. Super-Resolution with SRGAN.

MRI images are made even more spectacular with a Super- resolution Generative Adversarial Network (SRGAN). SRGAN is used to boost the resolution of MRI images by training sparse structural patterns using low-resolution images, which boosts the visual clarity of the image and helps to segment the tumor more accurately.

This is a result of the high resolution of images that enhances the clarity of groups of tumors and as a result, it causes a rise in the accuracy of segmentation. Other detection and segmentation models are then used to detect and segment the enhanced images using SRGAN.

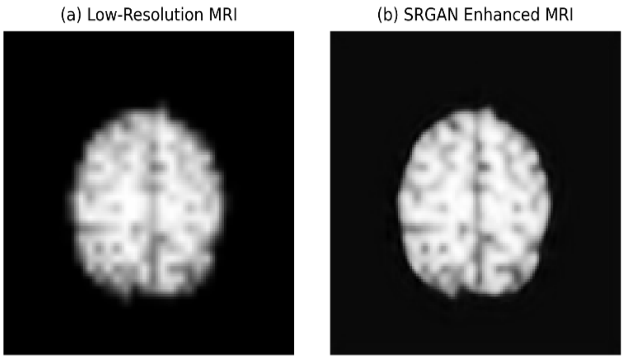

Fig. 2. Visual Comparison of the loa image to SRGAN high image using IXI Dataset

C. Deep Learning Model to Tumor Detection and Segmentation.

In this framework, there are a number of Deep Learning architectures applied to identify and localize tumors. Convolutional Neural Networks (CNN) is used with respect to learning sequential features of MRI images, to undertake feature engineering and categorization. They adopt encoder-decoder models such as U-Net so as to gain precise pixel-level segmentation. U-Net also contains skip connections which preserve spatial information hence the localisation of the tumor regions can be performed with accuracy.

In order to improve the functioning of segmentation, Attention U- Net is implemented and it incorporates attention unit to focus on the significant part of the tumor and deemphasize on the other nonrelevant background characteristics. This is so that it becomes easy to make the attention-guided learning that helps to capture the minute boundaries of the tumor especially when it is irregular shaped and low contrast in nature.

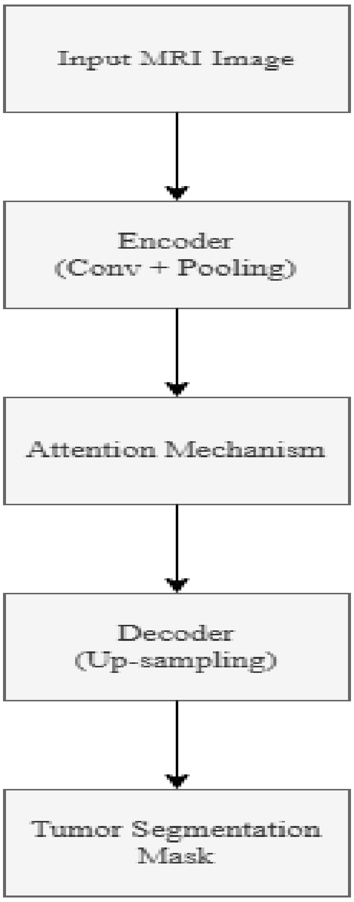

Fig. 3. Architecture of proposed Attention U-Net Model in the case of brain tumor Detection

D. Output Generation

The final product of the research proposal is segmented using MRI images where the spots of the tumor are readily defined. The outputs assist the radiologists to visualize the location, size and shape of the tumors and hence assist in proper diagnosis and planning of treatment procedures. The utility of the provided methodology is supported with the assistance of quantitative measures of evaluation which are discussed in the following sections.

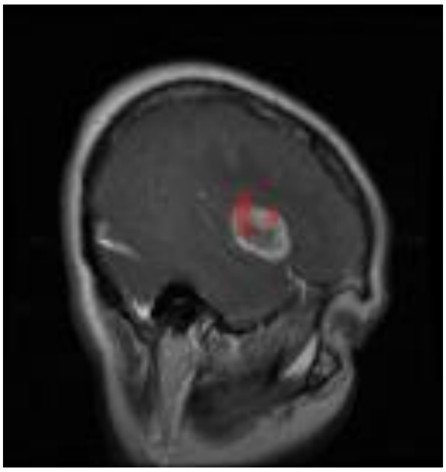

Fig. 4. This is output generation for the detection of the brain tumor with the segmented of the tumor marked in red mark.

IV. RESULT AND PERFORMANCE EVALUATION

Standard evaluation criteria are used to judge the performance of a segmentation task objectively in the proposed system. To determine how well the final cancer category is predicted, the accuracy criterion is employed because it estimates the proportion of correctly classified instances among the total number of instances. In addition, to check the overlap between the predicted region of the tumor and the expected region, overlap-based evaluation methods such as the Dice Similarity Coefficient or Dice Score, as well as the Intersection over Union metric, are used. Both metrics are widely used for the assessment of the medical image analysis task because they are successful methods used to judge the correctness of the boundary region.

The Dice Score is calculated as the doubled overlapping region between the output and the ground truth, and is normalized by the total number of pixels in the output and the ground truth. The IoU evaluates a ratio of the overlap region to the combination of both areas. The Dice Score value as well as IoU value will increase for greater segmentation accuracy and similarity with the ground truth. In the proposed system, a Convolutional Neural Network (CNN) auxiliary classifier will help in classifying the brain tumor as present or absent, while segmentation networks will help in locating the brain tumor region within the Magnetic Resonance Imaging (MRI) image. Therefore, the proposed system will result in a robust and reliable system for brain tumor diagnosis.

A. Tumor classification performance on Brain Tumor MRI Datasets

More specifically, the category of No Tumor, the model glorifies a better value of the Recall metric, which clearly shows the efficiency of the model in determining the normal images from the MRI images, as there will be fewer false negative values. Moreover, the balance between a high value of both Precision and Recall values determined in the Pituitary tumor category clearly shows the high-performing levels of the model at the tumor level. Despite the fact that fewer values were determined in the aspects of Recall in the Glioma and Meningioma categories, the satisfactory values of both F1 Scores clearly show that the high-performing levels of the proposed CNN model will remain stable.

Class-wise Performance of the Proposed Model

Class	Precision	Recall	F1-Score
Glioma	0.97	0.86	0.91
Meningioma	0.84	0.81	0.83
No Tumor	0.89	0.99	0.94
Pituitary	0.95	0.95	0.95

Table I. The following is the summary for the precision, recall, and F1 score for each class in the tumor classification model, which was able to attain an accuracy of 91%.

A.1 Performance Comparison of Brain Tumor Classes

The graphical representation illustrating the effect of the suggested CNN model for the classes compares of the proposed CNN model with other existing models for Precision, Recall, and the measure of F1-score for classes of the brain tumor. It has been observed that the model possesses a high level of precision for the Glioma classes and the Pituitary classes, which means the ability to identify tumor-specific MRI scans with less precision loss. The class 'No Tumor' possesses the highest measure of Recall, indicating the ability to not get classified as a tumor with a high level of precision. Even if the classes 'Meningioma' possess less precision and recall measures, the well-balanced measure of the classes illustrates the efficiency and potency of the proposed classification model.

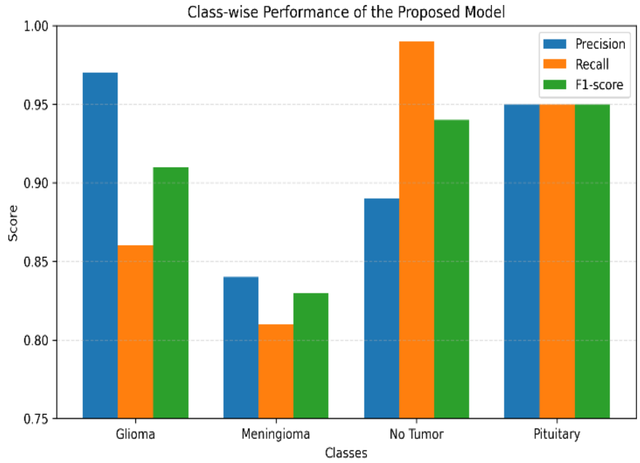

Fig. 5. Performance of the proposed CNN model class wise using precision, recall and F1-score for brain tumor classification

B. Tumor Segmentation Performance

The segmentation outcome illustrates the capability of the Attention U-Net architecture to perform better than general CNN and standard U-Net models. The use of attention mechanisms allows the network to better attend to the regions of interest, leading to more accurate location and definition of the boundaries and complex structures of the tumour regions. This, in turn, results in an improvement in the Dice and IoU scores, indicating better alignment and consistency between the segmentation outcome and the ground truth maps. Subsequently, the use of SRGAN to improve image quality also affects the overall performance accuracy, providing more refined details about the tumour region and leading to an improvement in image resolution, which promotes accurate feature identification within low-quality MRI images.

Performance Comparison of MRI Tumor Segmentation Models

Model	Accuracy (%)	Dice Score	IoU
CNN-based (Tumor-MRI)	92.5	0.87	0.78
U-Net (BRATS 2020)	99.09	0.6	0.49
Attention U-Net	99.09	0.94	0.89

Table II: This table shows the comparison done for this study among classification, segmentation, and Super-resolution models on various datasets

B.1.Comparative Analysis of MRI Tumor Segmentation Techniques

The bar chart above illustrates the relative performance of different models of MRI images of brain tumor segmentation in terms of Accuracy, Dice Score, and Intersection over Union (IoU). Tumor-MRI Model On CNN has a relative level of accurate performance of 92.50 percent, which is helpful for general tumor segmentation. The U-Net that is trained using BRATS 2020 is much accurate in terms of output; however, the scores of both Dice Score and IoU are low, which generally establishes that this model is not effective in tumor boundary identification. In contrast to the above-mentioned models of tumor segmentation of MRI images of the brain, the Attention U-Net model is much higher for all evaluation scores due to its high scores of Dice Score and IOU, which generally reveal that it is more capable of targeting areas related to the tumor. All the above-mentioned scores generally define that this study is effective and distinctive within its related counterparts for its relative accuracy and trustworthiness for reliability.

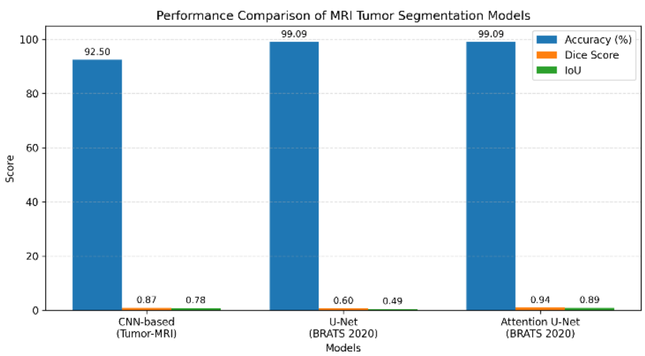

Fig. 6. Performance evaluation of MRI brain tumor segmentation using accuracy, Dice Similarity Coefficient, and Intersection over Union measures comparing various deep architectures.

C. Model Performance Comparison

We compared multiple deep learning models based on the Dice Score, and from that comparison, it was found that our proposed framework works effectively in segmenting brain tumors. It can be clearly observed from the graphical analysis that the proposed framework with SRGAN-based image enhancement and Attention U-Net yields higher normalized Dice scores throughout all training epochs compared with traditional CNN and standard U-Net models. Results justify that the incorporation of attention mechanisms significantly boosts the model's concentration on relevant tumor regions, thereby enabling its potential ability for accurate boundary localization. Furthermore, super-resolution enhancement enhances feature clarity and structural details in MRI images that facilitate faster convergence and improve the segmentation performance.

Overall, the graphical comparison clearly manifests the output of integrating attention and super-resolution techniques into superior and more stable segmentation accuracy, validating the effectiveness of the proposed approach.

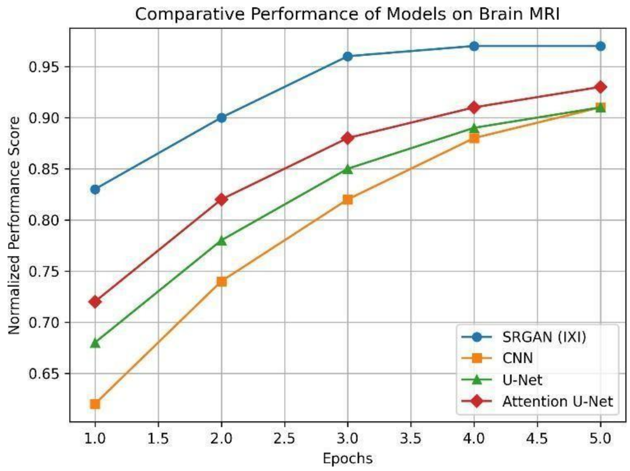

Fig. 7. Comparison of the performance of various deep learning models on the segmentation of the brain tumor using the normalized Dice similarity coefficient

D. Performance Comparison Analysis

I. Why Attention U-Net Performs Better

Attention U-Net enhances the performance of segmentation tasks with attention gates that enable the network to focus on the region that is relevant to tumors and overlook irrelevant information about the background, characteristics that are not of significance. This is relevant to images obtained from MRI for brains, where tumors appear to be small and irregular, and it helps in better demarcation and eradication of false positives. It is for this reason that Attention U-Net performs better than a regular U-Net in terms of Dice and IoU metrics.

II. Why CNN is Limited for Tumor Classification

Attention U-Net enhances the performance of segmentation tasks with attention gates that enable the network to focus on the region that is relevant to tumors and overlook irrelevant information about the background, characteristics that are not of significance. This is relevant to images obtained from MRI for brains, where tumors appear to be small and irregular, and it helps in better demarcation and eradication of false positives. It is for this reason that Attention U-Net performs better than of CNN model in terms of Dice and IoU metrics.

III. Why U-Net Performs Effectively for Brain Tumor Segmentation

The reason why the U-Net is applicable to Brain Tumor Segmentation is that it has a structure that is composed of both encoder and decoder with a skip connection, therefore conserving space and allowing for correct pixel-wise segmentation and accurate tumor localization. This is why the U-Net has become a strong benchmark for medical image segmentation.

IV. Why SRGAN is Useful Before Segementation

SRGAN improves the low-resolution images acquired from the MRI to better the structural edges and provide the lost high- frequency information. Before the stage of image segmentation, there is the potential to achieve improved image quality. feature extraction, hence promoting optimal tumor location and image segmentation results in the U-Net and Attention U-Net networks.

E. Qualitative Analysis

The quality of the proposed Deep Learning framework has been analyzed graphically to determine the significance of the proposed framework in locating as well as segregating the brain tumors from the images of the MRI scan. The difference between the segmentation result produced by the proposed framework and the actual true markings in the evaluation process would determine the level of accuracy in locating the tumor.

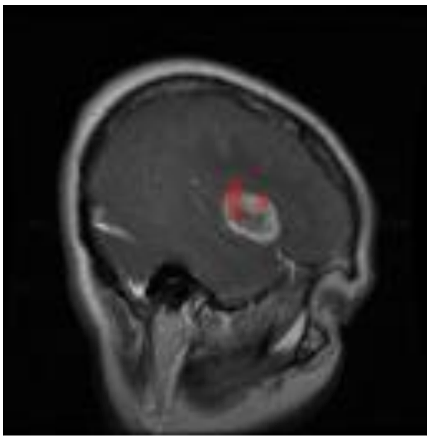

Fig. 8. Qualitative results demonstrating the original MRI image, ground truth tumor mask, as well as the segmentation results produced by our proposed Attention U-Net-based method.

It is further noted that the proposed model can efficiently eliminate the unwanted background region and target the actual tumor region. In this way, the effectiveness of the attention mechanism to promote the model for better image segmentation is manifested. Furthermore, the image enhancement performed by the SRGAN helps to boost the contrast of the tumor image.

It is clear that the qualitative analysis findings show that the solution using deep learning is able to ensure accurate and precise segmentation of the tumor that can be effectively used in a computer-aided solution for assisting a doctor or a radiologist in making a clinical decision.

V. CONCLUSION

The proposed article discusses an automated framework based on Using deep learning to find and separate brain tumors in MRI images. The proposed method includes the integration of image preprocessing and SRGAN-based super resolution enhancement and then relies on an advanced image segmentation method based on the Attention U-Net model and the CNN-based auxiliary classification technique for tumor detection. The reason why the model stays concentrated on the region-of-interest tumor area while filtering away the unnecessary information from the background is the attention mechanism.

Experimentations carried out on the proposed approach have proved its superiority over the conventional CNN approach and the standard U-Network Considering the Dice Similarity Coefficient and IoU scores. And the robustness, reliability, and effectiveness of the proposed system have also been asserted through the adoption of quantitative measures for assessment. Overall, the proposed approach proved its immense potential for implementation within the CAD systems for assisting the radiologists in accurate tumor localization.

VI. FUTURE WORK

This framework can be extended to handle multi-modal images of MRI like T1, T2, FLAIR, and contrast images to better characterize tumors so that the marking process can be carried out in an accurate manner. The transformer model, hybrid models, or whichever models are more appropriate should be applied to have better comprehension of features so that efficiency of the system can be increased.

In addition, the process of implementation of the model with respect to applications within the clinical environment may also be investigated to achieve a faster diagnosis. The integration of an AI-based graphical technology may help the radiologists to improve the interpretability of the result by giving a graphical representation of the diagnosis. In addition, the framework needs to be validated with more valid information to improve the level of generalisability.

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